
Experiment 6: Operational Amplifiers

ME-330

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1 Learning objective

The objectives of this experiment are to provide the student with an opportunity to:

- Become more familiar using Matlab for data acquisition, analysis, and dissemination (plotting, etc.).
- Design, build, and test a non-inverting amplifier circuit using a generic operational amplifier chip.
- Design, build, and test an instrumentation amplifier circuit using an instrumentation amplifier chip.
- Explore and verify the performance characteristics of operational amplifiers.
- Verify the gains of the constructed op-amp circuits.

2 Equipment checklist

- LM358AN Op-Amp
- AD622AN Instrumentation Amp
- Digilent Breadboard
- Powered Breadboard (or similar power supply)
- Wire & Resistor Kits
- Screw Driver, Pliers, Wire Stripper
- NI MyDAQ
- Analog Input (AI) and Analog Output (AO) on MyDAQ

3 Introduction

In this experiment you will construct and test two op-amp circuits. The first will be the non-inverting amplifier shown in Figure 1, using National Instruments' LM358 DIP (dual in-line package) chip. The second will be an instrumentation amplifier circuit using Analog Device's AD622 DIP. Both circuits will have a gain of approximately $G = 3$. Details for both of these DIP chips are given at the end of this document.

4 Laboratory instruction

The goal of this experiment is to learn about operational amplifier and setting up the gain in the op-amp circuit.

5 Part-A: Construct and test a non-inverting amplifier

- Fabricate a non-inverting amplifier on a breadboard. Select resistors from the supply kit at your station.
- Select $R_1 = 10\text{ k}\Omega$ and $R_2 = 20\text{ k}\Omega$ resistors from your resistor kit. Verify the correct color bands using a resistor chart (or internet resource), then measure and record the actual resistor values using the tabletop DMM in order to calculate the expected gain.
- Reference the LM358 datasheet for how to connect the DIP chip.
- Place the op-amp in the bread board across the center channel as seen in Figure 1.

- Construct your circuit on the breadboard in a neat and organized manner. Use pre-bent wires when possible; trim wires so that they are tight to the breadboard. Use the long vertical rails for your common signals (such as the +15V, -15V, ground, etc.). Your lab assistant may ask you to organize your wiring before helping you to debug your circuit.
- Use the NI MyDAQ power supply to provide +15V and -15V to the chip at the V+ and GND pins respectively (note – the GND pin on the LM358 amplifier is not the signal ground, it should be -15V for this experiment).
- Use the ground from the Digilent breadboard to ground the op-amp circuit.
- Connect your resistors as shown in Figure 1, with $R1 = 10\text{ k}\Omega$ and $R2 = 20\text{ k}\Omega$.
- Use the Analog Out (AO 0) on the NI MyDAQ for supplying the input voltage to your op-amp circuit (use a range of 0-5 Volts for input). This will be done in the Acquisition code provided for this experiment (see line 33 - 36 in the provided Matlab code for acquisition)
- Use the handheld digital multi-meter to test the non-inverting op-amp circuit to verify that it is performing correctly.
- Sample breadboard circuit is shown in Figure 1. Few things to note in the circuit. The wiring provided is color coded for clarity
 - Yellow color wires: are ground
 - Blue color wires: are V_{supply}^- negative supply voltage
 - Red color wires: are V_{supply}^+ negative supply voltage
 - Green color wire: are V_{out} output voltage signal, , Acquired at AI-0 of NI MyDAQ
 - Orange color wire: are V_{in} , input voltage signal, generated from AO-0 of NI MyDAQ and then Acquired at AI-1 of NI MyDAQ to record the data.
- Show your circuit to TAs before acquiring the data. **Wrong wire will damage the op-amp.**
- After the acquisition is done and data is recorded, **please use DMM to record input voltage and output voltage to the LM358 circuit in your log book.** *For check : The input voltage value should be close to 5 volts. The output voltage should be greater than 10 volts.*

6 Part-B: Create a Matlab Program to Acquire and Display Data from the NI MyDAQ.

You will the a generic acquisition program to collect the input to Op-Amp and output voltage from Op-Amp, see the sample code in the appendix 11.1.

- Make sure to set the sampling rate to 200 Hz and sampling to to 5 seconds.
- Make sure to acquire data on two channels of DAQ system. AI0 will be used for output signal from op-amp and AI1 will be used to acquire input voltage to op-amp.
- Make sure the output filename is correct before starting the acquisition program. Line 21 in the Matlab code.
- The program should gradually increase to 5 volts from 0 volts. This is done on the Analog output terminal “0”, it generates the input signal to the op-amp circuit. This will be done in the Acquisition code provided for this experiment (see line 33 - 36 in the provided Matlab code for acquisition)
- The acquisition program should acquire the data and store it in a file. Make sure the relative file path is correct.

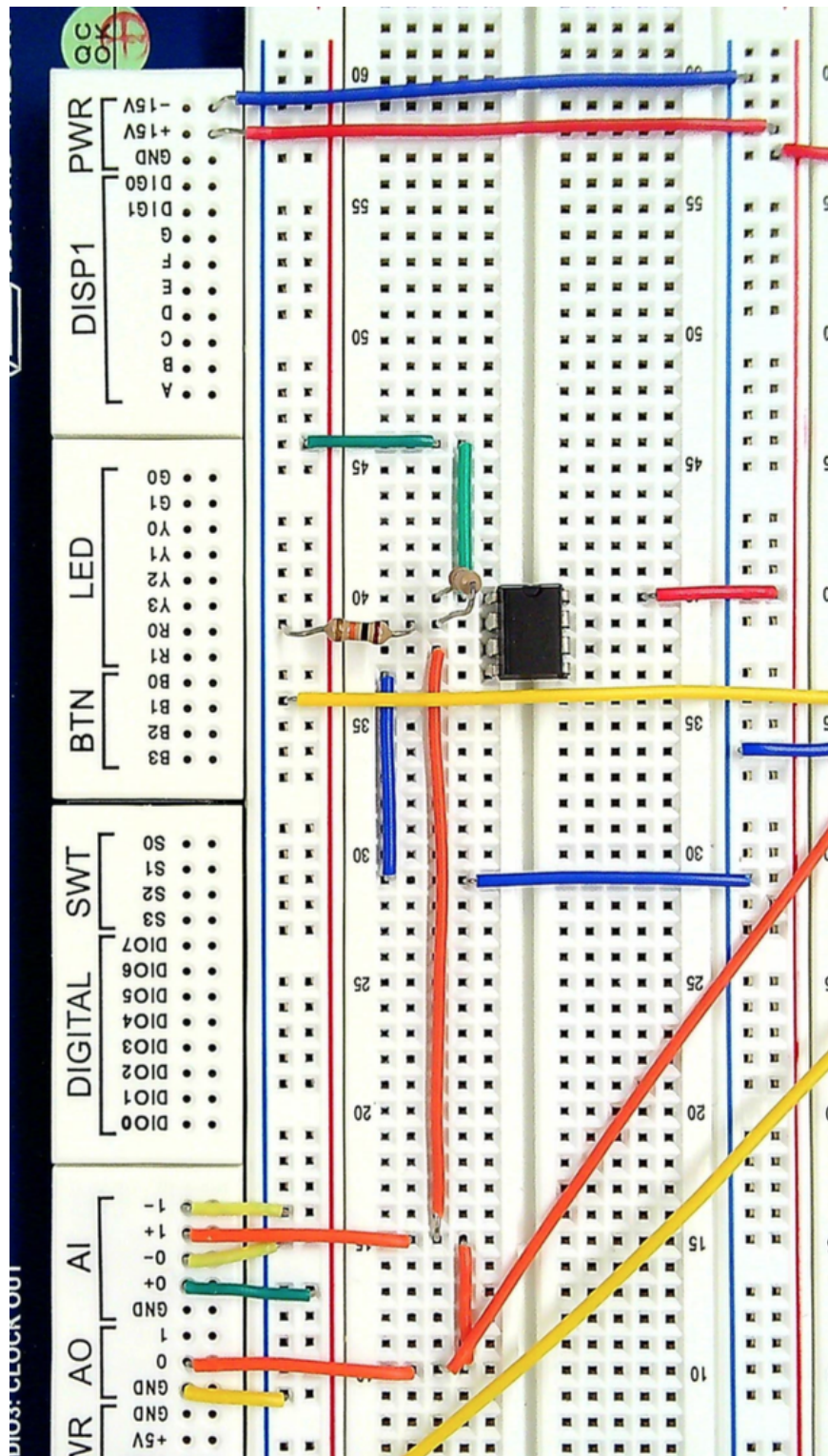


Figure 1: Sample wiring diagram for LM358 op-amp.

- For this part of the experiment the filename should be “LM358Data.dat”
- After the acquisition is done and data is recorded, **please use DMM to record input voltage and output voltage to the LM358 circuit in your log book.** *For check : The input voltage value should be close to 5 volts. The output voltage should be greater than 10 volts.*
- Why the recorded output voltage in log book is different than the output voltage shown in your DAQ plot?

7 Part-C: Connect and test an instrumentation amplifier

- Use the other side of the breadboard to build the circuit for AD622 chip.
- Construct an instrumentation amplifier with a gain of $G = 3$ on the breadboard. Select resistors from the supply provided at your station. The gain for the AD622 is set with an external resistor (use the datasheet link given at the end of this document as reference).
- Measure and record the actual resistor values.
- Construct your circuit on the breadboard in a neat and organized manner. Use pre-bent wires when possible; trim wires so that they are tight to the breadboard. Use the long vertical rails for your common signals (such as the +15V, -15V, ground, etc). If necessary, your lab assistant may ask you to organize your wiring before helping you to debug your circuit.
- Use the NI MyDAQ power on breadboard to provide +15V and -15V to the chip at the V+ and V- pins respectively.
- Use the ground from the Digilent breadboard to ground the circuit and the output ground from the op-amp circuit, as well as the REF pin on the instrumentation amplifier (this acts as signal ground).
- Use the Analog Out (AO 0) on the NI MyDAQ for supplying the input voltage to your op-amp circuit (use a range of 0-5 Volts for input). This will be done in the Acquisition code provided for this experiment (see line 33 - 36 in the provided Matlab code for acquisition)
- Use the handheld digital multi-meter to test the non-inverting op-amp circuit to verify that it is performing correctly.
- Sample breadboard circuit is shown in Figure 2.
- Show your circuit to TAs before switching on the power. **Wrong wire will damage the instrumentation amplifier.**
- Sample breadboard circuit is shown in Figure 1. Few things to note in the circuit. The wiring provided is color coded for clarity
 - Yellow color wires: are ground
 - Blue color wires: are V_{supply}^- negative supply voltage
 - Red color wires: are V_{supply}^+ positive supply voltage
 - Green color wire: are V_{out} output voltage signal, , Acquired at AI-0 of NI MyDAQ
 - Orange color wire: are V_{in} , input voltage signal, generated from AO-0 of NI MyDAQ and then Acquired at AI-1 of NI MyDAQ to record the data.
- Show your circuit to TAs before acquiring the data. **Wrong wire will damage the op-amp**
- After the acquisition is done and data is recorded, **please use DMM to record input voltage and output voltage to the AD622 circuit in your log book.** *For check : The input voltage value should be close to 5 volts. The output voltage should be greater than 10 volts.*

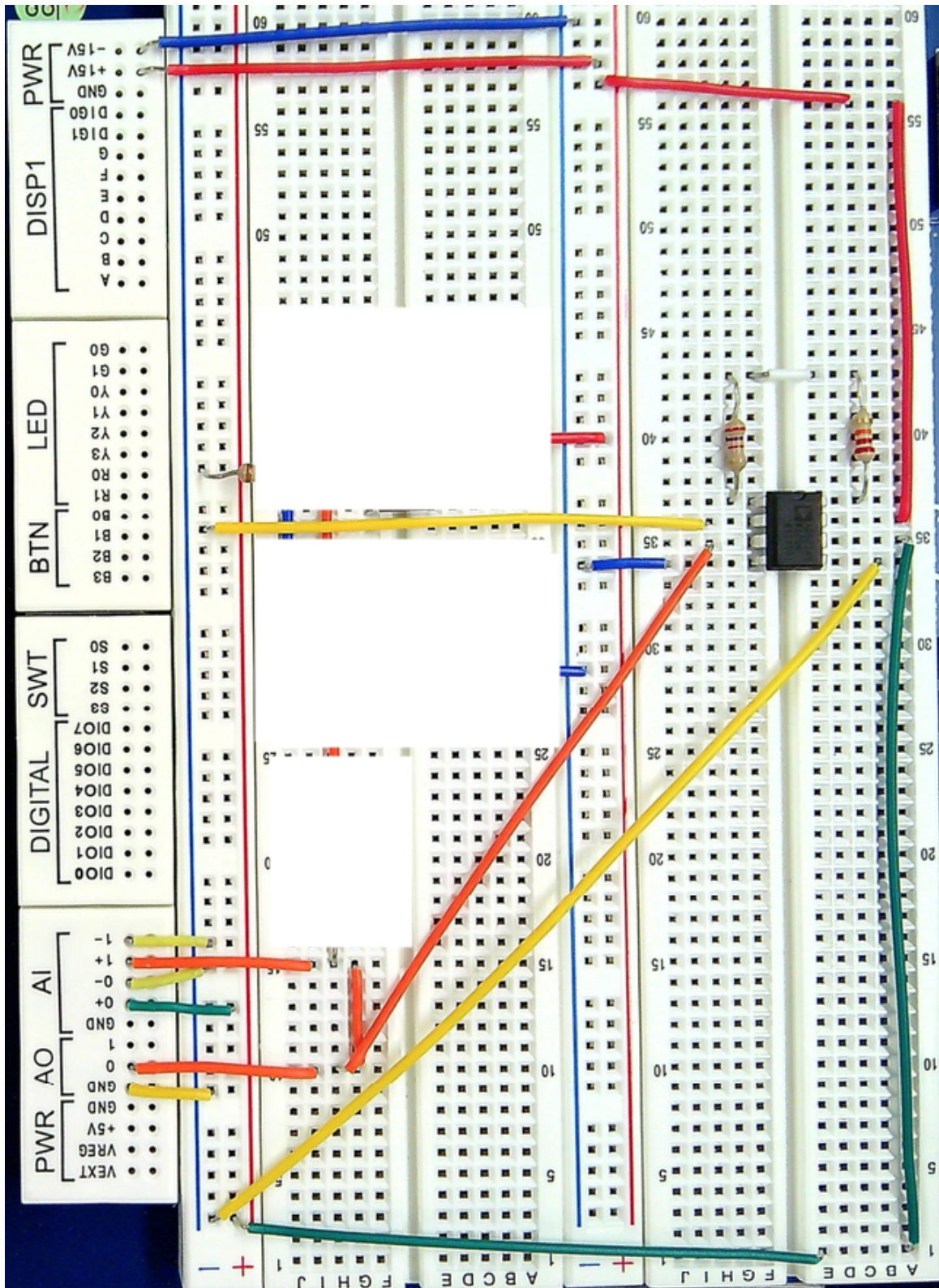


Figure 2: Sample wiring diagram for AD622 instrument amplifier.

8 Part-D: Create a Matlab Program to Acquire and Display Data from the NI MyDAQ. Code provided for this semester

You will the a generic acquisition program to collect the input to Op-Amp and output voltage from Op-Amp, see the sample code in the appendix 11.1.

- Make sure to set the sampling rate to 200 Hz and sampling to to 5 seconds.
- Make sure to acquire data on two channels of DAQ system. AI0 will be used for output signal from op-amp and AI1 will be used to acquire input voltage to AD622.
- Make sure the output filename is correct before starting the acquisition program. Line 21 in the Matlab code.
- The program should gradually increase the input signal voltage to 5 volts from 0 volts. This is done on the Analog output terminal “0”, it generates the input signal. This will be done in the Acquisition code provided for this experiment (see line 33 - 36 in the provided Matlab code for acquisition)
- The acquisition program should acquire the data and store it in a file. Make sure the relative file path is correct.
- For this part of the experiment the filename should be “LM358Data.dat”
- After the acquisition is done and data is recorded, **please use DMM to record input voltage and output voltage to the AD622 circuit in your log book.** *For check : The input voltage value should be close to 5 volts. The output voltage should be greater than 10 volts.*
- Why the recorded output voltage in log book is different than the output voltage shown in your DAQ plot?

9 Part-F

Return lab space to prior condition

- Turn off the power supply.
- Remove wires connecting the power supply to the breadboard and return to the wall
- Remove all breadboard wires and place them back in the wire kit in an organized fashion.
- Remove the connecting wires from the breadboard and set aside.
- Log off the computer.
- Make sure to copy the data to your harddrive.
- **Do this for every lab!**

10 Post-Laboratory

For this experiment, submit the following items on Canvas for your post-laboratory assignment.

- Amplifier gain plot. Submit a single figure with a plot showing the output vs. input data and the linear regression line from both amplifier circuits. Sample plot is shown in Figure 3. Specifically, the single plot should include:
 - Output vs. Input data from the LM358 op-amp circuit, including saturation (top Plot). Use markers.

- Linear regression line from the linear portion of the output vs. input data from the LM358 circuit.
 - Output vs. Input data from the AD622 instrumentation amplifier circuit, including saturation. Use markers (bottom plot).
 - Linear regression line from the linear portion of the output vs. input data from the AD622 instrumentation amplifier circuit. This plot should also show output saturation when the circuit output nears the supply voltage limits. Make sure to properly annotate the plot.
 - Sample Matlab code for plotting is shown in the appendix [11.2](#)
- Answer all question in the post-lab assessment on Canvas.

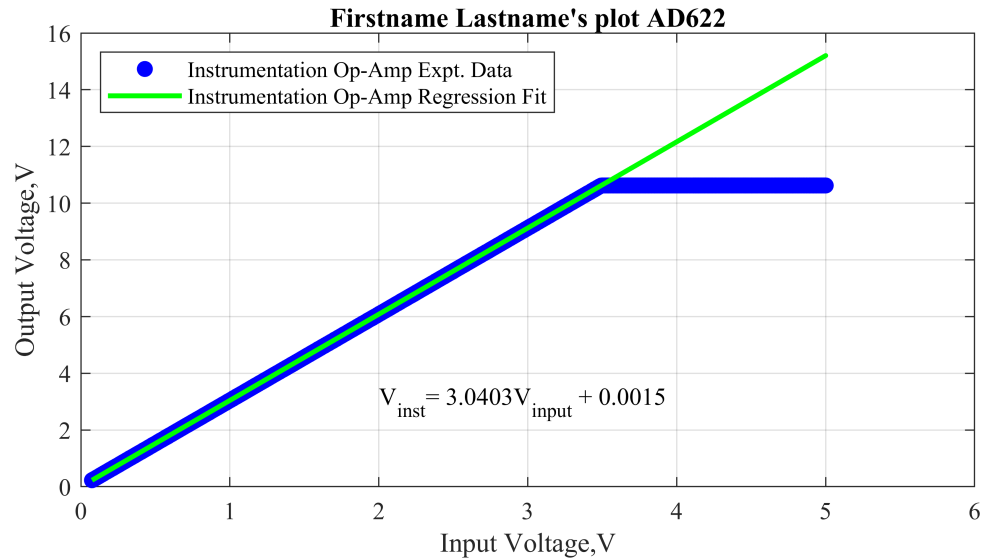
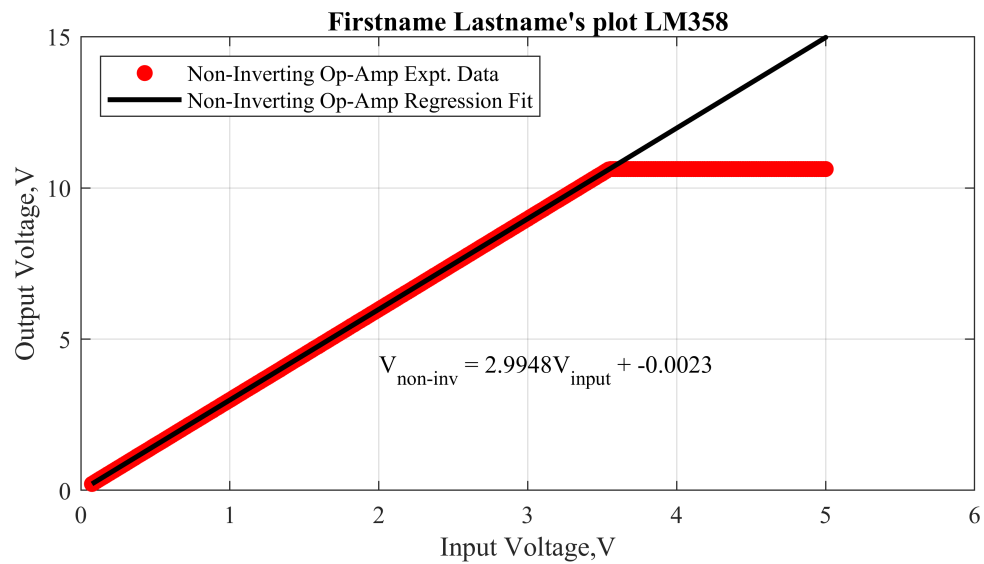


Figure 3: Sample plot for post-lab.

11 Appendix

11.1 Matlab program to acquire data from op-amp

```
1 % This is a generic program to acquire the data from NI- 6001
2 %% Sample data acquisition program for collecting the data for OP-AMP circuit
3 % file for later use
4 % Date: July 19th
5 % Dr. Vibhav Durgesh
6 % Rev 0.0
7 % This is basic program to acquire data from a data acquisition device.
8 % User has to provide appropriate information see - beginning of the code
9 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
10 % THIS CODE OVERWRITE THE DATA. PLEASE MOVE THE FILES OR REMANE PRIOR TO
11 % RERUNNING THE CODE
12 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
13
14 clear all
15 close all
16 clc
17
18 %% ----- START USER REQUIRED INFORMATION ----- %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
19 Fs = 200; %Sampling rate Data per sec
20 T = 5; % Time to acquire data
21 filename = '..\Data\AD622Data.dat'; % To store experiment data %% Change this name to
    appropriate chip
22 fid1 = fopen(filename,'w');
23
24 plotTitle = 'Firstname Lastname''s plot - Experiment#6';
25 fprintf(fid1,'%s \t %s \t %s\n','time (s)', 'Output Voltage(v)', 'Input Voltage(v)');
26 % -----END USER REQUIRED INFORMATION ----- %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
27 d = daq.getDevices;
28 s = daq.createSession('ni');
29 ch = addAnalogInputChannel(s,'MyDAQ1', [0 1], 'Voltage'); % Acquiring data on 2 channels
30 s.Rate = Fs; % Sampling rate modify as required
31 s.DurationInSeconds = T; % Sampling time modify as required
32
33 addAnalogOutputChannel(s,'MyDAQ1',0,'Voltage');
34 s.Rate = Fs;
35 outputSignal1 = linspace(0,5,Fs*T)';
36 queueOutputData(s,outputSignal1);
37
38 lh1 = addlistener(s,'DataAvailable', @(src, event)plotAndLogData(src, event));
39 lh2 = addlistener(s,'DataAvailable', @(src, event)logData(src, event, fid1));
40 errorListener = addlistener(s, 'ErrorOccurred', @(src,event) disp(getReport(event.Error)));
41 drawnow
42 startForeground(s);
43 delete(s)
44 delete(lh1)
45 delete(lh2)
46 fclose(fid1)
47 title(plotTitle,'fontname','times','fontsize',14)
48
49 function plotAndLogData(src,event)
50 time = event.TimeStamps;
51 InputData = event.Data(:,1);
52 OutputData = event.Data(:,2);
53 figure(1)
54 plot(time, InputData,'k-');hold on
55 plot(time, OutputData,'r-');
56 xlabel('time (s)')
57 ylabel('Voltage(v)')
58 legend('Input volt','Output volt','location','northwest')
59 grid on
60 end
61
62 function logData(src, evt, fid)
```

```
63 % Add the time stamp and the data values to data. To write data sequentially,  
64 % transpose the matrix.  
65 data = [evt.TimeStamps evt.Data(:,1) evt.Data(:,2)] ;  
66 fprintf(fid,'%3.8f \t %3.8f \t %3.8f \n',data');  
67 end
```

11.2 Matlab program to plot op-amp response

```
1 % This is a generic program to plot the data acquired from the op-amp and
2 % instrument amplifier
3 % Date: July 19th
4 % Dr. Vibhav Durgesh
5 % Rev 0.0
6 % Rev 0.1 Modified for new DAQ and circuit.
7 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
8 clear all
9 close all
10 clc
11
12
13 fid = fopen('..\Data\LM358Data.dat'); % Input data file for LM358
14 line1 = fgetl(fid);
15 data1 = fscanf(fid,'%f \n', [3 inf]);
16 time1 = data1(1,:);
17 output1 = data1(2,:);
18 input1 = data1(3,:);
19 fclose(fid);
20
21 fid = fopen('..\Data\AD622Data.dat'); % Input data file for AD622
22 line1 = fgetl(fid);
23 data2 = fscanf(fid,'%f \n', [3 inf]);
24 time2 = data2(1,:);
25 output2 = data2(2,:);
26 input2 = data2(3,:);
27 fclose(fid);
28 nstr = 1;
29 nend = 600;
30 %% Perform Linear Regression for Non-Inverting Amplifier
31 [p,s] = polyfit(input1(nstr:nend),output1(nstr:nend),1);
32 xfitLM358 = input1;
33 yfitLM358 = polyval(p,xfitLM358);
34
35 [p2,s2] = polyfit(input2(nstr:nend),output2(nstr:nend),1);
36 xfitAD622 = input2;
37 yfitAD622 = polyval(p2,xfitAD622);
38
39
40 % Generating figure with specific size
41 figure(1)
42 set(gcf,'unit','inches','position',[0.50 0.50 6.50 7.50],...
43 'defaultaxesfontsize',10,'defaultaxesfontname','times');
44 % Plotting data
45 subplot(2,1,1)
46 plot(input1(1:3:end),output1(1:3:end),'ro','markersize',6,'markerfacecolor','r');hold on
47 plot(xfitLM358,yfitLM358,'k-','linewidth',2)
48 %xlim([0 3.5])
49 %ylim([0 10])
50 xlabel('Input Voltage,V')
51 ylabel('Output Voltage,V')
52 legend('Non-Inverting Op-Amp Expt. Data','Non-Inverting Op-Amp Regression Fit','location','
53 northwest')
54 text(2,4,sprintf('V_{non-inv} = %3.4f%s + %3.4f',p(1),'V_{input}',p(2)),'Fontname','times')
55 grid on
56 title('Firstname Lastname''s plot LM358')
57 subplot(2,1,2)
58 plot(input2,output2,'bo','markersize',6,'markerfacecolor','b'); hold on;
59 plot(xfitAD622,yfitAD622,'g-','linewidth',2)
60 %xlim([0 3.5])
61 %ylim([0 10])
62 xlabel('Input Voltage,V')
63 ylabel('Output Voltage,V')
64 legend('Instrumentation Op-Amp Expt. Data','Instrumentation Op-Amp Regression Fit','location
65 ', 'northwest')
66 text(2,3,sprintf('V_{inst}= %3.4f%s + %3.4f',p2(1),'V_{input}',p2(2)),'Fontname','times')
```



```

65 title('Firstname Lastname''s plot AD622')
66 grid on
67 % % % Saving the files in png and pdf format
68 figName = ['..\Figures\Firstname_Lastname_Exp06_Part1'];
69 set(gcf,'PaperPositionMode','auto')
70 print(figName,'-dpng','-r600')
71 set(gcf,'PaperUnits','inches','Units','inches');
72 figpos = get(gcf,'Position');
73 set(gcf,'PaperSize',figpos(3:4),'Units','inches');
74 print(figName,'-dpdf','-r600')

```

11.3 OP-AMP LM358

The LM358 DIP (dual in-line package) actually includes two internal operational amplifiers. For this experiment you will be using only 1 side (A or B) of the chip. A connection diagram is given in Figure 4. To determine which side is which, locate the half circle indentation on the top of the chip and match it to the diagram. The chip must be powered by an external ± 12 or ± 15 Volt external power supply (for this lab we will use the powered breadboard or similar). The +15 Volt source should be connected to V+ pin and the -15 Volt source should be connected to the GND pin. Please note that the GND pin should not be connected to the signal ground! Make sure, however, that the ground from the input signal is connected to the ground from the powered breadboard, and that you measure the output voltage from pin 1 or pin 7 (depending on which side you use) to the common ground (not the GND pin!). A detailed datasheet for the LM358N can be found at: [Sparkfun website](#).

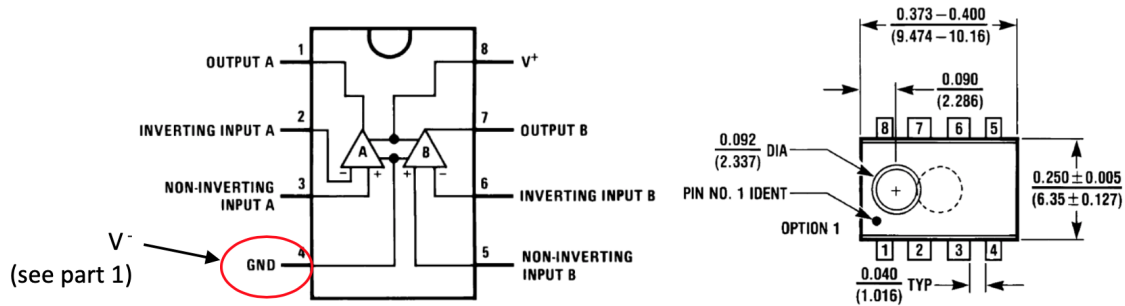


Figure 4: Connection diagram and package outline for the LM358 operational amplifier.

11.4 Instrument amplifier AD-622

The AD622 DIP (dual in-line package) contains several operational amplifiers arranged in an instrumentation amplifier circuit utilizing precision resistors in order to improve the accuracy of the gain and to reduce uncertainty. A connection diagram is shown in Figure 5. Refer to the datasheet below for determining how to change the gain through the external resistor R_G . The AD622 datasheet can be found at: [AD622 Webpage](#).

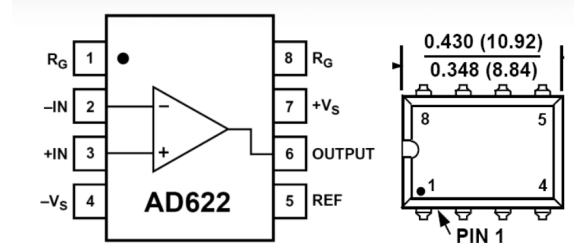


Figure 5: Connection diagram and package outline for the AD622 instrumentation amplifier.

The AD622 gain is resistor programmed by R_G or, more precisely, by whatever impedance appears between Pin 1 and Pin 8. The AD622 is designed to offer gains as close as possible to popular integer values using standard 1% resistors. Figure 6 shows required values of R_G for various gains. Note that for $G = 1$, the R_G pins are unconnected ($R_G = \infty$). For any arbitrary gain, R_G can be calculated by using the formula

$$R_G = \frac{50.5k\Omega}{G - 1} \quad (1)$$

To minimize gain error, avoid high parasitic resistance in series with R_G . To minimize gain drift, R_G should have a low temperature coefficient less than 10 ppm/ $^{\circ}\text{C}$ for the best performance.

Desired Gain	1% Std Table Value of R_G , Ω	Calculated Gain
2	51.1 k	1.988
5	12.7 k	4.976
10	5.62 k	9.986
20	2.67 k	19.91
33	1.58 k	32.96
40	1.3 k	39.85
50	1.02 k	50.50
65	787	65.17
100	511	99.83
200	255	199.0
500	102	496.1
1000	51.1	989.3

Figure 6: Required Values of Gain Resistor.